

Step 6: Front-End

BCS3210 Blockchains group project: Decentralized Digital Identity and Data Sharing Platform

What we did in this step

We made wireframes for the four screens and then built a small web front-end ("ID Share") with plain HTML, CSS and JavaScript. It talks to the contracts on the local Hardhat node with **viem** (the same library as in the Hardhat scripts). With it you can register an identity, give and revoke consent, request access and read the audit log. We tested every function by hand in the browser.

1. Wireframes

We kept the layout simple: an account selector at the top, four tabs (Patient, Requester, Admin, Audit log) and cards with one task each.

1. Patient: register (not registered yet)

2. Patient: identity and consents

3. Requester: request access

4. Admin and audit log

2. How it works

Part	What it does
frontend/index.html, style.css	Page layout with the four tabs; the grid falls back to one column on small screens
frontend/app.js	viem publicClient for reads and events, walletClient for transactions, both over HTTP to http://127.0.0.1:8545

Part	What it does
frontend/contracts.js	Addresses and ABIs, generated after deployment by scripts/export-frontend.mjs from the Ignition output and the Hardhat artifacts
frontend/server.mjs	Tiny static file server (Node, no extra packages) on http://localhost:5173

- **Accounts.** For the demo the page uses the default Hardhat test accounts (Admin, Patient 1, Doctor, Stranger, Patient 2, Lab). viem signs the transactions locally with these public test keys, so no wallet extension is needed. In a real version this would be MetaMask.
- **Registration.** The patient picks the two ID photos. The browser computes the SHA-256 hashes (crypto.subtle) and the salted email hash (keccak256(encodePacked(salt, email))), so the photos never leave the computer. The salt is shown once so the patient can keep it.
- **Lists** (approved requesters, patients, consents, a requester's own requests) are built from contract events, for example ConsentGranted filtered by patient, plus a call to getConsent for the current state.
- **Errors.** When a transaction reverts, the require message from the contract is shown, for example "Duration must be 1-365 days".

Start it:

```
npx hardhat node          # terminal 1
npm run deploy:local      # terminal 2
npm run frontend          # then open http://localhost:5173
```

3. Screenshots

Patient: registration form (account that is not registered yet)

ID Share

Account

Patient 2 - 0x15d3...6A65

0x15d34AAf54267DB7D7c367839AAf71A00a2C6A65 | 0 ACT

Patient

Requester

Admin

Audit log

Register your identity

The photos stay on your computer. Only their SHA-256 hashes go on-chain.

Email

patient@example.test

Storage reference

vault://patient1

Front of ID

Choose File No file chosen

Back of ID

Choose File No file chosen

Hash and register

My identity

Nothing yet

Give consent

Requester

Doctor (0x3C44...93BC)

What can they see

Front and back

My consents

Nothing yet

Patient: identity, give consent, list of consents with revoke

ID Share

Account

Patient 1 - 0x7099...79C8

0x70997970C51812dc3A010C7d01b50e0d17dc79C8 | 20 ACT

Patient

Requester

Admin

Audit log

My identity

Field	Value
Email hash	0xe288d12b4a68961d7d837a9803f90fe0834832581244adb5f324208993a72f83
Front photo hash	0xcbbfde808d9adce00f84b08b05f42aebef3ab306f3e77f895880476f92a9bb3b
Back photo hash	0x2058e8c5667f7d5dd251e5546446567fc89540028210eda67dfe6e6cd8c441b3
Storage reference	vault://patient1
Registered	9/30/2026, 3:16:56 PM

My consents

Requester	Scope	Valid until	State	
Doctor (0x3C44...93BC)	Front and back	10/30/2026, 2:17:04 PM	Active	Revoke
Lab (0x9965...A4dc)	Front only	10/7/2026, 3:17:16 PM	Revoked	

Give consent

Requester

Doctor (0x3C44...93BC)

What can they see

Front and back

Days (1-365)

30

Grant consent

Requester: request access and own request history

ID Share

Account

Doctor - 0x3C44...93BC

0x3C44CdDdB6a900fa2b585dd299e03d12FA4293BC | 0 ACT

Patient

Requester

Admin

Audit log

Request access

Every request is written to the patient's audit log, also when it is denied.

Patient

Patient 1 (0x7099...79C8)

Request access

My requests

Block	Patient	Result	Details
11	Patient 1 (0x7099...79C8)	GRANTED	Front and back

Admin: approve / remove requesters, pause

ID Share

Account
Admin - 0xf39F...2266
0xf39Fd6e51aad88F6F4ce6aB8827279cFfFb92266 | 0 ACT

Patient

Requester

Admin

Audit log

Approve requester

Address
0x...

Approve

Platform

Data sharing: **running**

PauseUnpause

Reward per first consent: **10** ACT

Approved requesters

Requester

Doctor (0x3C44...93BC)Remove

Lab (0x9965...A4dc)Remove

Audit log of a patient (newest first, granted and denied attempts)

ID Share

Account
Patient 1 - 0x7099...79C8
0x70997970C51812dc3A010C7d01b50e0d17dc79C8 | 20 ACT

Patient

Requester

Admin

Audit log

Audit log

Patient
Patient 1 (0x7099...79C8)

#	Time	Requester	Result	Details
2	9/30/2026, 3:18:04 PM	Lab (0x9965...A4dc)	DENIED	Consent revoked
1	9/30/2026, 3:17:43 PM	Stranger (0x90F7...b906)	DENIED	Requester not approved
0	9/30/2026, 3:17:32 PM	Doctor (0x3C44...93BC)	GRANTED	Front and back

4. Manual test in the browser

On a fresh node we clicked through this scenario; everything behaved as expected and there were no errors in the browser console.

Step	Action in the UI	Result
1	Admin approves Doctor and Lab	Both listed as approved (49,692 gas each)
2	Patient 1 registers with two photos	Hashes shown under "My identity" (161,311 gas)
3	Patient 1 gives Doctor "front and back" for 30 days, Lab "front only" for 7 days	Both consents Active, balance 20 ACT
4	Patient 1 tries 400 days	Error "Duration must be 1-365 days", nothing changed
5	Doctor requests access	GRANTED, tx hash shown for the vault, Doctor still has 0 ACT
6	Stranger requests access	DENIED, "Requester not approved", logged
7	Patient 1 revokes Lab, Lab requests	DENIED, "Consent revoked", logged
8	Admin pauses, Doctor requests, Admin unpauses	"Contract is paused" error, then running again
9	Audit log tab for Patient 1	Shows the 3 logged attempts with results and reasons
10	Phone width (375 px)	One column layout, no horizontal scrolling